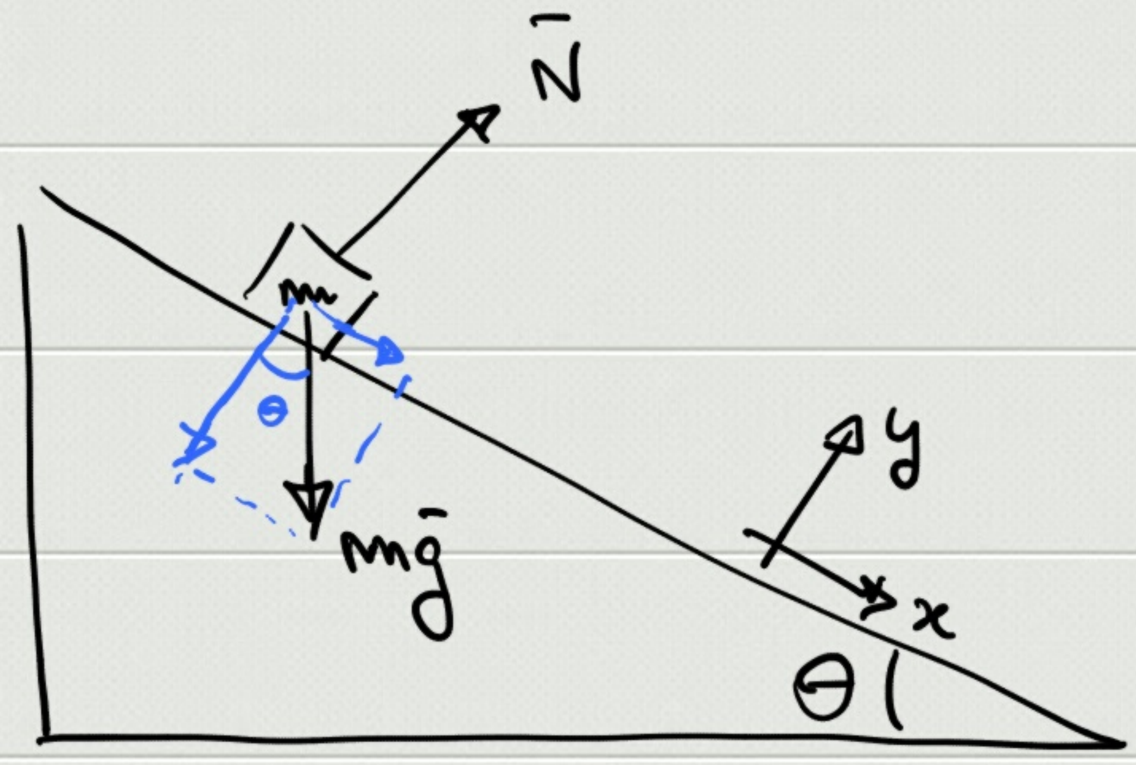


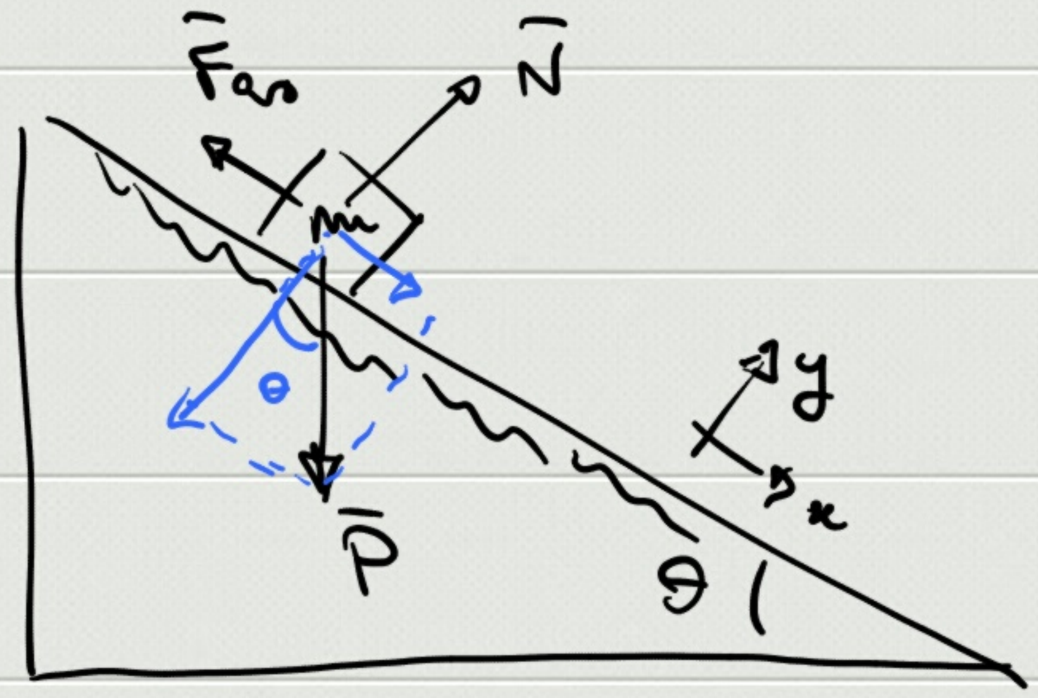
$$\sum_i \bar{F}_i = m\bar{a}$$

$\boxed{\text{liscio}}$

$$m\bar{g} + \bar{N} = m\bar{a}$$

$x: \ \cancel{m}g\sin\theta = \cancel{m}a_x \quad \Rightarrow \boxed{a = a_x = g\sin\theta}$

$y: \ -mg\cos\theta + N = ma_y = 0 \quad \Rightarrow N = mg\cos\theta$



Statica

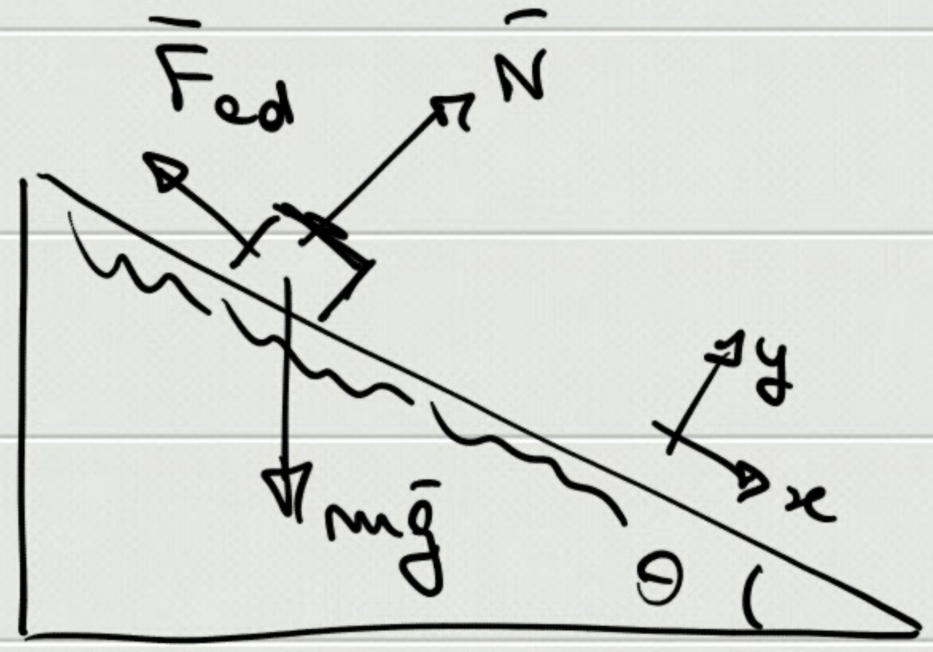
$$\bar{F}_{as} + \bar{N} + \bar{P} = 0$$

$$x: \quad -F_{as} + mg\sin\theta = 0 \quad \Rightarrow F_{as} = mg\sin\theta$$

$$y: \quad N - mg\cos\theta = 0 \quad \Rightarrow N = mg\cos\theta$$

$$\left.\begin{aligned} F_{as} \le F_{as,max} &= \mu_s N = \mu_s \cancel{m}g\cos\theta \\ &= \cancel{m}g\sin\theta \end{aligned}\right\} \Rightarrow$$

$$\boxed{\mu_s \ge tg\,\theta}$$



Dinamica

$$\bar{F}_{ed} + \bar{N} + \bar{P} = m\bar{a}$$

$$x: \quad -\mu_d N + mg\sin\theta = ma$$

$$y: \quad N - mg\cos\theta = 0 \implies N = mg\cos\theta$$

$$\cancel{m}g(\sin\theta - \mu_d\cos\theta) = \cancel{m}a$$

$$\Rightarrow \boxed{a = g(\sin\theta - \mu_d\cos\theta)}$$